SmartApti Test

Student Assessment Result

Student USN: 4JN25CS175

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Test: Class 2 Part2 17092026

20 / 20

Score: 100%

Correct: 20 **Wrong: 0** **Unanswered: 0**

Answer Review

Q1. Which of the following best describes a computer?

Your Answer: B) An electronic device that accepts data, processes it, and produces information

Correct Option: B) An electronic device that accepts data, processes it, and produces information

Explanation: A computer accepts input data, processes it according to instructions, stores data when required, and produces useful output.

Q2. Which generation of computers primarily used vacuum tubes?

Your Answer: A) First generation

Correct Option: A) First generation

Explanation: First-generation computers used vacuum tubes for electronic processing. They were large, generated considerable heat, and consumed substantial power.

Q3. Which technology was a major feature of second-generation computers?

Your Answer: B) Transistors

Correct Option: B) Transistors

Explanation: Second-generation computers replaced vacuum tubes with transistors, making computers smaller, faster, more reliable, and more energy-efficient.

Q4. Which of the following is NOT generally classified as a class of computer based on size or processing capability?

Your Answer: D) Keyboard computer

Correct Option: D) Keyboard computer

Explanation: Supercomputers, mainframes, and minicomputers are recognized computer classes. A keyboard is an input device, not a class of computer.

Q5. Which type of computer is generally designed to perform extremely large numbers of calculations at very high speed?

Your Answer: A) Supercomputer

Correct Option: A) Supercomputer

Explanation: Supercomputers are built for highly demanding computational tasks such as scientific simulations, weather modeling, and complex numerical analysis. Page 2

Q6. A computer used by a single person for everyday tasks such as word processing and web browsing is commonly called a:

Your Answer: C) Personal computer

Correct Option: C) Personal computer

Explanation: A personal computer (PC) is intended for individual use and commonly supports applications such as word processing, browsing, programming, and multimedia.

Q7. Which memory unit is the smallest among the following?

Your Answer: B) Bit

Correct Option: B) Bit

Explanation: A bit is the smallest basic unit of digital information and can represent either 0 or

Q8. How many bits are contained in one byte?

Your Answer: C) 8

Correct Option: C) 8

Explanation: One byte consists of 8 bits. A bit represents a binary digit, while a byte is commonly used as a basic unit for representing data.

Q9. Which of the following is a volatile memory?

Your Answer: B) RAM

Correct Option: B) RAM

Explanation: RAM is volatile memory, meaning its contents are normally lost when power is removed. It temporarily holds programs and data currently being used.

Q10. Which memory is normally used to store firmware that must be retained when power is switched off?

Your Answer: C) ROM

Correct Option: C) ROM

Explanation: ROM is non-volatile memory and can retain stored information without continuous power. It is commonly associated with firmware or boot-related instructions. Page 3

Q11. The main purpose of the ALU is to:

Your Answer: B) Perform arithmetic and logical operations

Correct Option: B) Perform arithmetic and logical operations

Explanation: The Arithmetic Logic Unit (ALU) performs arithmetic operations such as addition and subtraction and logical operations such as comparisons and Boolean operations.

Q12. Which of the following operations is normally performed by the ALU?

Your Answer: A) Addition

Correct Option: A) Addition

Explanation: Addition is an arithmetic operation and is therefore performed by the ALU. The other activities are handled by peripheral devices or physical connections.

Q13. What is the primary role of the Control Unit (CU) in a processor?

Your Answer: B) It directs and coordinates the execution of instructions

Correct Option: B) It directs and coordinates the execution of instructions

Explanation: The Control Unit fetches and interprets instructions and coordinates the activities of the processor and other components during instruction execution.

Q14. Which register holds the address of the memory location that is to be accessed?

Your Answer: A) MAR

Correct Option: A) MAR

Explanation: The Memory Address Register (MAR) stores the address of the memory location involved in a read or write operation.

Q15. Which register temporarily holds data being transferred to or from memory?

Your Answer: B) MDR

Correct Option: B) MDR

Explanation: The Memory Data Register (MDR) holds data being transferred between the processor and memory during memory read or write operations. Page 4

Q16. Which register normally contains the address of the next instruction to be fetched?

Your Answer: C) PC

Correct Option: C) PC

Explanation: The Program Counter (PC) keeps track of the address of the next instruction to be fetched for execution, subject to changes caused by branches and other control-flow operations.

Q17. Which register stores the instruction currently being decoded or executed?

Your Answer: A) IR

Correct Option: A) IR

Explanation: The Instruction Register (IR) holds the current instruction after it has been fetched from memory so that the control unit can decode it.

Q18. In the processor-memory connection shown in the figure, which component provides the address used to identify a memory location?

Your Answer: A) MAR

Correct Option: A) MAR

Explanation: The MAR provides the memory address. The memory uses that address to identify the location from which data is read or to which data is written.

Q19. Which statement about general-purpose registers is correct?

Your Answer: B) They can hold temporary data and intermediate results during processing

Correct Option: B) They can hold temporary data and intermediate results during processing

Explanation: General-purpose registers are fast storage locations inside the processor. They can hold operands, temporary values, addresses, and intermediate results during instruction execution.

Q20. Which sequence correctly represents a simplified instruction-fetch process?

Your Answer: C) Fetch !’ Decode !’ Execute

Correct Option: C) Fetch !’ Decode !’ Execute

Explanation: In the basic instruction cycle, the processor fetches an instruction, decodes what it means, and then executes the required operation.